

An exact causal audit for action conditioned world models

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Status: Technical research note with a complete proof for a synthetic finite family and a reproducible exact checker. Novelty has not been established. This is not a peer reviewed paper or a solution to general world model learning.

Abstract

Can perfect action conditioned prediction and positive frequency of every logged action certify good planning? Without a causal identification assumption, they cannot. We give an explicit family of one step environments in which every action appears equally often, the complete logged data law is identical across environments, and a reward predictor has zero logged prediction error. Nevertheless, the exact minimax expected deployment regret from those logs is $(1 - 1/m)^2$ for m actions. We derive exact minimax regret for both a balanced intervention design and an optimal adaptive design with a fixed query budget, assuming independent hidden state resets. The analysis separates ambiguity, decision error, and expected regret. A dependency free checker verifies the finite examples using rational arithmetic, independently enumerated latent and observable outcomes, and Bellman optimization over all query actions. The broad identification obstruction is established prior knowledge; the contribution here is a fully specified, auditable teaching and evaluation fixture, with no priority claim for the construction or formulas.

1 Motivation and relationship to prior work

World models are often evaluated on held out trajectories generated by a logging policy, then used to compare interventions generated by a planner. These are different statistical questions when the logging policy uses information missing from the data available to the planner.

Tennenholtz, Mannor, and Shalit study off policy evaluation under partial observability, explain failures of standard estimators, and give identification results under additional structure [1]. Pace and colleagues explicitly study nonidentifiable hidden confounding through multiple world models compatible with the same observations [2]. Thus, neither hidden confounding nor observationally equivalent models with different policy values is a new idea introduced here.

Grimm and colleagues formalize value equivalence for planning [3]. A recent preprint by Wei and colleagues distinguishes mechanism, response, and decision sufficiency, and observes in Section 3.2 that information requirements alone do not guarantee that observations identify the required distinction [4]. This note isolates that latter obstacle in a finite, exactly solvable example. It does not establish a new representation learning method, identify a flaw in those papers, or evaluate a deployed world model.

The ongoing broader challenge is to justify decision quality from affordable world model evaluations under stated observability, data collection, and deployment assumptions. The example below supplies one necessary negative control for that challenge. It does not solve the general challenge.

2 Environment and information available to the learner

Fix an integer $m \geq 2$. There are m actions $\mathcal{A} = \{1, \dots, m\}$ and m possible environments, indexed by a fixed but unknown $j \in \mathcal{A}$. Each episode draws a fresh hidden variable U

uniformly from $\mathcal{A}$. The preaction observation is a constant; the deployment agent does not observe U .

The reward in environment j is

$$R_j(U, a) = \begin{cases} 1/2, & a = U, \\ 1, & a \neq U \text{ and } a = j, \\ 0, & a \neq U \text{ and } a \neq j. \end{cases}$$

All rewards lie in $[0, 1]$. The privileged logging policy observes U and chooses $A = U$. The recorded variables are the constant observation, A , and R ; U is not provided to the learner. Logged episodes are independent. At deployment, the learner chooses one action using its logged dataset and private randomization, before seeing a fresh independent U .

The learner is allowed to know the entire family and its data collection mechanism. It does not know j . There is no computational or model capacity restriction. Giving the learner this structural knowledge makes the impossibility claim stronger, and is essential to the interpretation of the later optimal audit rule.

Write

$$V_j(a) = \mathbb{E}_U[R_j(U, a)], \quad \Delta = 1 - \frac{1}{m}.$$

For a randomized action rule π and a dataset D , expected regret is

$$\text{Reg}_j(\pi) = \mathbb{E}_{D, \pi} \left[\max_a V_j(a) - V_j(A) \right].$$

This is a one step partially observed controlled environment with privileged data collection. It is a special case of a planning problem. It is not a claim about all partially observed systems: in particular, behavior policies measurable only from the fully recorded observation history are outside this counterexample's privileged logging assumption.

3 Perfect logged prediction with an exact regret obstruction

Proposition 1. For every j , the logged law is

$$\Pr_j(A = a, R = 1/2) = 1/m \quad \text{for every } a.$$

The predictor that assigns probability one to reward $1/2$ for every action has zero logged squared error and zero logarithmic loss on the discrete reward alphabet. Nevertheless,

$$\inf_{\pi} \max_{j \in \mathcal{A}} \text{Reg}_j(\pi) = \left(1 - \frac{1}{m}\right)^2. \quad (1)$$

The infimum is over all learners based only on any amount of these logs, including randomized learners.

Proof. Because $A = U$, every observed reward is $1/2$. The action is uniform and its distribution is independent of j . Hence the complete law of any finite logged dataset is identical in every environment; observing its population law also reveals no j .

Under intervention, by contrast,

$$V_j(j) = 1 - \frac{1}{2m}, \quad V_j(a) = \frac{1}{2m} \quad (a \neq j).$$

The unique optimal action is j , and its gap to every other action is Δ . Let $p_a = \Pr(A = a)$ be the learner's marginal deployment action probabilities, averaging over its logs and internal randomness. Since its inputs have the same law in every environment, the vector p is independent of j . Therefore

$$\text{Reg}_j(\pi) = \Delta(1 - p_j), \quad \max_j \text{Reg}_j(\pi) = \Delta(1 - \min_j p_j) \geq \Delta(1 - 1/m) = \Delta^2.$$

Choosing uniformly among all actions achieves equality for every j . This proves (1). $\square$

As the number of actions increases, this worst case regret tends to the largest possible reward range. The quantifier matters: for every learner there is a compatible environment with this regret. The statement does not say that every learner fails in every environment.

The observed action coverage is genuine but insufficient: $\Pr(A = a) = 1/m > 0$. Conditional on the hidden state, however, the logger has zero probability of every action except U . Conditioning on an observed action selects $U = a$, while intervening on that action leaves U uniform. More passive samples or a larger predictive network cannot remove this distinction in the specified family.

4 An exact balanced intervention audit

Suppose the learner may run a fixed audit: it executes each action n times, where $n \geq 0$ is an integer. Every query uses a fresh independent uniform U , independent of the chosen action and all previous queries. The action schedule may be deterministic. It is the independence from U , not randomness of the schedule alone, that makes these queries interventions.

The audit records all rewards. It then chooses a deployment action. This section optimizes that final decision rule for the fixed balanced design; it does not optimize over adaptive query designs or over all designs with the same budget.

For an action equal to j , reward 1 identifies the optimal action. For any other action, reward 0 rules it out. Reward 1/2 is uninformative about whether that action is j . If no reward 1 occurred, define the surviving set S to contain actions that never produced reward 0. The decision rule selects the action identified by a 1, if present, and otherwise chooses uniformly from S .

Set $q = m^{-n}$, the probability that an action's entire audit block consists of rewards 1/2.

Proposition 2. For the known finite family and fixed balanced reset audit, the exact mini-max expected regret is

$$\mathcal{R}(m, n) = \left(1 - \frac{1}{m}\right) \left[q - \frac{1 - (1 - q)^m}{m} \right]. \quad (2)$$

The probability that the audit leaves more than one compatible environment is

$$\mathcal{A}(m, n) = q [1 - (1 - q)^{m-1}]. \quad (3)$$

Equation (3) is an ambiguity or abstention probability. It is not the error probability of a decision rule, which may select correctly even when several environments remain compatible.

Proof. Under true environment j , its block fails to produce a 1 with probability q . Conditional on this event, each of the other $m - 1$ actions independently survives with probability q . Consequently

$$|S| = 1 + B, \quad B \sim \text{Binomial}(m - 1, q).$$

The rule's regret on this event is $\Delta B/(1 + B)$; otherwise it is zero. Its expected regret is thus $\Delta q \mathbb{E}[B/(1 + B)]$. The elementary identity

$$\mathbb{E} \left[\frac{1}{1 + B} \right] = \sum_{b=0}^{m-1} \frac{1}{b + 1} \binom{m-1}{b} q^b (1 - q)^{m-1-b} = \frac{1 - (1 - q)^m}{mq}$$

gives (2). Ambiguity occurs precisely when the j block is uninformative and $B \geq 1$, which gives (3).

To establish minimax optimality, place a uniform prior on j . If a 1 is observed, only its action can be j . Otherwise the likelihood of the entire transcript is equal for every $j \in S$ and zero for $j \notin S$: each surviving action has an all-1/2 block, and each eliminated action has the same nonoptimal block law in every surviving world. Therefore the posterior on S is uniform. Selecting any action in S is Bayes optimal. The proposed uniform choice is symmetric across world labels, so its frequentist risk is the same for every j . Its maximal risk equals its Bayes risk, which lower bounds the maximal risk of every rule. It is therefore minimax. $\square$

Useful special cases and a bound are

$$\mathcal{R}(m, 0) = \left(1 - \frac{1}{m}\right)^2, \quad \mathcal{R}(m, 1) = \frac{(1 - 1/m)^{m+1}}{m},$$

$$\mathcal{R}(m, n) \leq \frac{(1 - 1/m)(m - 1)}{2} m^{-2n}.$$

The last inequality follows from $B/(1 + B) \leq B/2$ for every nonnegative integer B . These formulas assume independent hidden state resets for all audit queries. They do not apply unchanged to paired branches sharing a hidden state, temporally correlated resets, measurement noise, or unknown environment families.

4.1 Optimal adaptive queries within the same family

We can also optimize the query design, still within this exact known family. The learner has a budget of b reset queries, may adapt each action to previous audit rewards, and then chooses one deployment action. Audit rewards and query costs do not enter the objective; only the expected regret of the final deployment decision is minimized. The fixed world index j persists, while every query and deployment episode has a fresh independent U .

Proposition 3. Let $B_b \sim \text{Binomial}(b, \Delta)$ and $x_+ = \max(x, 0)$. The minimax expected deployment regret over all such adaptive audit and decision rules is

$$\mathcal{R}_{\text{adapt}}(m, b) = \frac{\Delta}{m} \mathbb{E}[(m - 1 - B_b)_+]. \quad (4)$$

One minimax optimal rule chooses each query uniformly from the still compatible actions until a reward 1 identifies j or only one compatible action remains. It ignores uninformative rewards $1/2$, eliminates an action after its reward 0, and chooses uniformly among compatible actions when the budget ends.

Proof. Under a uniform prior on j , the posterior remains uniform on its surviving set after every audit history. An adaptive policy's probability of selecting its next action is a function of the observed history, so that factor is the same in each surviving world and does not invalidate the posterior calculation.

Suppose k candidates survive and b queries remain. Write $G(k, b)$ for the optimal Bayes expected deployment regret. Its boundary values are

$$G(k, 0) = \Delta \frac{k-1}{k}, \quad G(1, b) = 0.$$

A query of an eliminated action provides no information about j . All surviving actions are equivalent under the uniform posterior. Querying one gives an uninformative reward with probability $1/m$, identifies j with probability Δ/k , and eliminates one candidate with probability $\Delta(k-1)/k$. Information can be ignored, so querying a survivor is never worse than querying an eliminated action. Thus, for $k \geq 2$ and $b \geq 1$, the exact optimality recurrence is

$$G(k, b) = \frac{1}{m} G(k, b-1) + \Delta \frac{k-1}{k} G(k-1, b-1). \quad (5)$$

Induction on b , using the binomial recursion, gives

$$G(k, b) = \frac{\Delta}{k} \mathbb{E}[(k-1 - B_b)_+].$$

This establishes the Bayes lower bound at $k = m$. A strategy that chooses query and final actions symmetrically among survivors has the same risk for every world label, so its maximal risk equals the Bayes lower bound. This proves minimax equality (4). $\square$

For $b = m$, identity $\mathbb{E}[(m-1 - B_m)_+] = \Delta^m$ shows

$$\mathcal{R}_{\text{adapt}}(m, m) = \Delta^{m+1}/m = \mathcal{R}(m, 1).$$

Thus one balanced sweep is already optimal for that query budget in this family. At other budgets adaptivity need not improve strictly; for example, with two actions both formulas at budget $b = 2n$ equal 2^{-2n-2} . These are properties of the specified erasure-like observation channel, not general claims about world model audits.

5 Exact computational verification

Run from this directory using Python 3.10 or later and only its standard library:

```
python verify_causal_audit.py --output exact_results.json --check-report
↪ report.md
```

The checker constructs the reward tables directly, verifies equality of logged laws, and compares actual interventional values with the formulas. For the smaller audit instances it enumerates every latent state sequence and independently checks the same transcript probabilities by multiplying outcome multiplicities. For each resulting transcript it checks the feasible worlds, equal posterior likelihoods, and regret of the decision rule from the actual reward table. Additional balanced audit cases use the factorized route only. The adaptive check derives all posterior transitions directly from the reward table and minimizes over every action by exact Bellman recursion, including actions already ruled out. The exact test scope is recorded in `exact_results.json`; it is finite corroboration of the proof, not a substitute for the proof or a formal proof assistant certificate.

The following table is generated from exact results and checked byte for byte on each release verification run.

Actions m	Queries per action n	Total audit queries	Balanced design regret	Optimal adaptive regret at same budget	Balanced ambiguity probability
2	0	0	1/4	1/4	1
2	1	2	1/16	1/16	1/4
4	0	0	9/16	9/16	1
4	1	4	243/4096	243/4096	37/256
4	2	8	4419/1048576	909/1048576	721/65536
5	0	0	16/25	16/25	1
5	1	5	4096/78125	4096/78125	369/3125

An initial exact run preceded the written verification plan. The release run follows the fixed acceptance criteria in `verification_plan.md`. This chronology is disclosed to avoid presenting the work as a preregistered empirical discovery. There is no Monte Carlo sampling, training run, checkpoint, benchmark dataset, or statistical significance test.

6 Evaluation implications and limits

An evaluation intended to support causal planning should specify what the data collector observed, what was recorded, and why action assignment is independent of unrecorded reward relevant variables under the stated conditioning. Merely showing every action frequently, or reporting a strong conditional prediction score, does not establish this property.

For this fixture, the audit report should separate logged fit, support based ambiguity, and deployment regret. An implementation that reports successful causal identification from the passive logs has failed the fixture even if its prediction score is perfect. The exact audit formula is an oracle reference for this family; it should not be used as an estimated guarantee for a neural world model without the family assumptions.

In the ordinary unconfounded case, conditional outcome distributions given actions can have a causal interpretation. For example, let O be recorded before the action, and let U include the relevant common causes of assignment and reward, with no remaining confounding. If A is conditionally independent of U given O , the reward mechanism is unchanged by assignment, and the action has positive conditional probability, then

$$\Pr(R \mid O, A = a) = \sum_u \Pr(R \mid O, U = u, a) \Pr(U = u \mid O) = \Pr(R \mid O, \text{do}(A = a)).$$

This elementary sufficient condition explains which assumption the construction violates. Partial observability alone is not an impossibility theorem.

No claim is made that this note advances the state of the art in causal identification, offline reinforcement learning, or general world model planning. Establishing novelty would require a substantially broader literature comparison. Moving beyond a public technical note would also require a clear novel question, stronger results or relevant model experiments, and independent expert review.

Claim ledger

Claim	Evidence and permitted interpretation
Logged prediction plus marginal action coverage need not identify interventions	Known principle, illustrated by Proposition 1; no novelty claim
Exact no-audit regret for the specified family	Complete proof of (1), finite exact checks
Exact fixed-design post-audit minimax regret	Complete proof of (2), finite exact checks; no optimal-design claim
Exact optimal adaptive audit regret	Complete proof of (4) within this family, plus finite exact Bellman optimization; not a general intervention design theorem
Exact ambiguity probability	Complete proof of (3); not interchangeable with decision error
Performance on trained world models or physical systems	Not tested; no empirical performance claim
General world-model problem solved, peer review, or new result priority	Not claimed

Authorship and AI assistance

潘奕成 (Yicheng Pan) directed the research topic and requested the public research artifact. An OpenAI assistant generated the initial construction, derivations, implementation, and draft prose. AI assisted checking covered the principal formulas and assumptions. These checks are not external expert validation. This note does not assert that the named author personally wrote the proof or code without AI assistance. Human expert review and a comprehensive novelty assessment have not been completed.

References

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3. Christopher Grimm, André Barreto, Satinder Singh, and David Silver. The Value Equivalence Principle for Model Based Reinforcement Learning. NeurIPS 2020. [Author preprint](#), [proceedings paper](#).
4. Rongzhe Wei, Hans Hao-Hsun Hsu, Peizhi Niu, Yifan Li, and Pan Li. What Must a World Model Distinguish for Planning? Preprint, September 2026. [Author preprint](#). Cited as a preprint, without implying peer review.

Primary sources checked on 2 October 2026. No source text or code is redistributed in this package.